

T3 UV and EFT Lagrangians

EFT order: 5; loop order: 1.

$$T3 - B, \quad \alpha = -1 : \quad S_1 = (2, -\frac{1}{2}), \quad S_2 = (2, \frac{1}{2}), \quad F = (1, 0)$$

$$\begin{aligned}
\text{Full UV Lagrangian} & -\frac{B_{\mu_1\mu_2}^2}{4g_Y^2} - \frac{\left(G_{\mu_1\mu_2}^{X_1}\right)^2}{4g_s^2} \\
& -\frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2} \\
& +i\overline{d_{\alpha_1,i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(d_{\alpha_1,i_1}) \\
& +i\overline{e_{i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(e_{i_1}) \\
& +i\overline{q_{\alpha_1,a_1,i_1}}\gamma^{\mu_1}P_LD_{\mu_1}(q_{\alpha_1,a_1,i_1}) \\
& +i\overline{u_{\alpha_1,i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_1}) \\
& +i\overline{\ell_{a_1,i_1}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_1}) \\
& +\frac{1}{2}iN_{r_1}{}^TC\gamma^{\mu_1}D_{\mu_1}(N_{r_1}) \\
& -\frac{1}{2}N_{r_1}{}^TCN_{r_1}M_F \\
& +\overline{\ell_{a_1,i_1}}P_RN_{r_1}\text{NewScalar1}_{a_1}y_{1,i_1,r_1} \\
& -\overline{q_{\alpha_1,a_1,i_1}}P_Rd_{\alpha_1,i_2}H_{a_1}Y_{d,i_1,i_2} \\
& -\overline{\ell_{a_1,i_1}}P_Re_{i_2}H_{a_1}Y_{e,i_1,i_2} \\
& -\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_2}\bar{e}_{a_1,a_2} \\
& +\overline{y_{2,i_1,r_1}}N_{r_1}{}^TC P_L\ell_{a_1,i_1}\text{NewScalar2}_{a_2}\bar{e}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_1}{}^\dagger \\
& -\overline{Y_{e,i_1,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}H_{a_1}{}^\dagger \\
& -\overline{q_{\alpha_1,a_2,i_1}}P_Ru_{\alpha_1,i_2}Y_{u,i_1,i_2}\epsilon_{a_2,a_1}H_{a_1}{}^\dagger \\
& +H_{a_1}\mu^2H_{a_1}{}^\dagger \\
& -\frac{1}{2}\lambda H_{a_1}H_{a_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger \\
& +\overline{y_{1,i_1,r_1}}N_{r_1}{}^TC P_L\ell_{a_1,i_1}\text{NewScalar1}_{a_1}{}^\dagger \\
& -M_{S_1}^2\text{NewScalar1}_{a_1}\text{NewScalar1}_{a_1}{}^\dagger \\
& +\text{lambdaH1Inv1}H_{a_3}\text{NewScalar1}_{a_4}\text{T3HiggsPortal1CGInv1}_{a_1,a_3,a_2,a_4}H_{a_1}{}^\dagger\text{NewScalar1}_{a_2}{}^\dagger \\
& +\text{lambdaH1Inv2}H_{a_3}\text{NewScalar1}_{a_4}\text{T3HiggsPortal1CGInv2}_{a_1,a_3,a_2,a_4}H_{a_1}{}^\dagger\text{NewScalar1}_{a_2}{}^\dagger \\
& +\frac{1}{2}\text{lambdaS1Inv1NewScalar1}_{a_3}\text{NewScalar1}_{a_4}\text{T3Scalar1SelfCGInv1}_{a_1,a_3,a_2,a_4}\text{NewScalar1}_{a_1}{}^\dagger\text{NewScalar1}_{a_2}{}^\dagger \\
& +\overline{\lambda_{T3}}\text{NewScalar2}_{a_1}\epsilon_{a_2,a_3}H_{a_1}{}^\dagger H_{a_2}{}^\dagger\text{NewScalar1}_{a_3}{}^\dagger \\
& -M_{S_2}^2\text{NewScalar2}_{a_1}\text{NewScalar2}_{a_1}{}^\dagger \\
& +\overline{\ell_{a_2,i_1}}P_RN_{r_1}y_{2,i_1,r_1}\epsilon_{a_2,a_1}\text{NewScalar2}_{a_1}{}^\dagger \\
& +H_{a_1}H_{a_2}\text{NewScalar1}_{a_3}\lambda_{T3}\bar{e}_{a_2,a_3}\text{NewScalar2}_{a_1}{}^\dagger \\
& +\text{lambdaH2Inv1}H_{a_3}\text{NewScalar2}_{a_4}\text{T3HiggsPortal2CGInv1}_{a_1,a_3,a_2,a_4}H_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +\text{lambdaH2Inv2}H_{a_3}\text{NewScalar2}_{a_4}\text{T3HiggsPortal2CGInv2}_{a_1,a_3,a_2,a_4}H_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +\text{lambda12Inv1NewScalar1}_{a_3}\text{NewScalar2}_{a_4}\text{T3CrossScalarCGInv1}_{a_1,a_3,a_2,a_4}\text{NewScalar1}_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +\text{lambda12Inv2NewScalar1}_{a_3}\text{NewScalar2}_{a_4}\text{T3CrossScalarCGInv2}_{a_1,a_3,a_2,a_4}\text{NewScalar1}_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +\frac{1}{2}\text{lambdaS2Inv1NewScalar2}_{a_3}\text{NewScalar2}_{a_4}\text{T3Scalar2SelfCGInv1}_{a_1,a_3,a_2,a_4}\text{NewScalar2}_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger
\end{aligned}$$

$$\begin{aligned}
& +D_{\mu_1}(H_{a_1})^\dagger D_{\mu_1}(H_{a_1}) \\
& +D_{\mu_1}(\text{NewScalar1}_{a_1})^\dagger D_{\mu_1}(\text{NewScalar1}_{a_1}) \\
& +D_{\mu_1}(\text{NewScalar2}_{a_1})^\dagger D_{\mu_1}(\text{NewScalar2}_{a_1}) \\
& \text{Matched EFT Lagrangian} - \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S2}^2)^2}
\end{aligned}$$

ω

$$\begin{aligned}
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{12 \epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{(G_{\mu_1 \mu_2}^{X_1})^2}{4 g_s^2}
\end{aligned}$$

$$\begin{aligned}
& -\frac{\hbar \left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{12 \epsilon} \\
& -\frac{1}{24} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2}\right) \left(W_{\mu_1 \mu_2}^{A_1}\right)^2 \\
& -\frac{1}{24} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_2}^2}\right) \left(W_{\mu_1 \mu_2}^{A_1}\right)^2 \\
& -\frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4 g_L^2} \\
& +i \overline{d_{\alpha_1, i_2}} \gamma^{\mu_1} P_R D_{\mu_1} \left(d_{\alpha_1, i_2}\right) \\
& +i \overline{e_{i_2}} \gamma^{\mu_1} P_R D_{\mu_1} \left(e_{i_2}\right) \\
& +i \overline{q_{\alpha_1, a_1, i_2}} \gamma^{\mu_1} P_L D_{\mu_1} \left(q_{\alpha_1, a_1, i_2}\right) \\
& +i \overline{u_{\alpha_1, i_2}} \gamma^{\mu_1} P_R D_{\mu_1} \left(u_{\alpha_1, i_2}\right) \\
& +i \overline{\ell_{a_1, i_2}} \gamma^{\mu_1} P_L D_{\mu_1} \left(\ell_{a_1, i_2}\right) \\
& -\overline{q_{\alpha_1, a_1, i_2}} P_R d_{\alpha_1, i_1} H_{a_1} Y_{d, i_2, i_1} \\
& -\overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} Y_{e, i_2, i_1} \\
& +\frac{\hbar y_{1, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1}}{4 \epsilon} \\
& -\frac{1}{8} \hbar y_{1, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} \\
& -\frac{\hbar y_{1, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_F^2 - M_{S_1}^2\right)} \\
& -\frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2}\right) \overline{y_{1, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_F^2 - M_{S_1}^2\right)} \\
& -\frac{\hbar y_{1, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_F^2 - M_{S_1}^2\right)^2} \\
& -\frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2}\right) \overline{y_{1, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_F^2 - M_{S_1}^2\right)^2} \\
& +\frac{\hbar y_{2, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1}}{4 \epsilon} \\
& -\frac{1}{8} \hbar y_{2, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} \\
& -\frac{\hbar y_{2, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_F^2 - M_{S_2}^2\right)} \\
& -\frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_2}^2}\right) \overline{y_{2, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_F^2 - M_{S_2}^2\right)} \\
& -\frac{\hbar y_{2, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_F^2 - M_{S_2}^2\right)^2} \\
& -\frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_2}^2}\right) \overline{y_{2, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_F^2 - M_{S_2}^2\right)^2}
\end{aligned}$$

$$\begin{aligned}
& -\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_2} \bar{\epsilon}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_1}^\dagger \\
& -\overline{Y_{e,i_1,i_2}}\overline{e_{i_2}} P_L \ell_{a_1,i_1} H_{a_1}^\dagger \\
& +C_{H2} H_{a_1} H_{a_1}^\dagger \\
& +\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& -\frac{1}{8} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& -\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S1}^2)} \\
& -\frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S1}^2)} \\
& -\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S1}^2)^2} \\
& -\frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S1}^2)^2} \\
& +\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& -\frac{1}{8} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& -\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S2}^2)} \\
& -\frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S2}^2)} \\
& -\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S2}^2)^2} \\
& -\frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S2}^2)^2} \\
& -\overline{q_{\alpha_1,a_2,i_2}} P_R u_{\alpha_1,i_1} Y_{u,i_2,i_1} \epsilon_{a_2,a_1} H_{a_1}^\dagger \\
& -\frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{\text{lambdaH1Inv}1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{5\text{lambdaH1Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& +\frac{\text{lambdaH2Inv}1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{5\text{lambdaH2Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& +\frac{\text{lambdaH1Inv}1\text{lambdaH1Inv}2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}
\end{aligned}$$

$\mathcal{L}$

$$\begin{aligned}
& + \frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{1}{2}\text{lambdaH1Inv1}^2 \hbar \log\left(\frac{\mu\text{bar2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6}\text{lambdaH1Inv2}^2 \hbar \log\left(\frac{\mu\text{bar2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH1Inv1}\text{lambdaH1Inv2}\hbar \log\left(\frac{\mu\text{bar2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{1}{2}\text{lambdaH2Inv1}^2 \hbar \log\left(\frac{\mu\text{bar2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6}\text{lambdaH2Inv2}^2 \hbar \log\left(\frac{\mu\text{bar2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar \log\left(\frac{\mu\text{bar2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& + \frac{\hbar \log\left(\frac{\mu\text{bar2}}{M_{S1}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& - \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& - \frac{\hbar \log\left(\frac{\mu\text{bar2}}{M_{S2}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2}
\end{aligned}$$

$$T3 - C, \; \alpha = -1 : \quad S_1 = (2, -\tfrac{1}{2}), \quad S_2 = (2, \tfrac{1}{2}), \quad F = (3, 0)$$

$$\begin{aligned}
& \text{Full UV Lagrangian} - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2} - \frac{\left(G_{\mu_1 \mu_2}^{X_1}\right)^2}{4g_S^2} \\
& - \frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4g_L^2} \\
& + i \overline{d_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1, i_1}) \\
& + i \overline{e_{i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_1}) \\
& + i \overline{q_{\alpha_1, a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1, a_1, i_1})
\end{aligned}$$

∞

$$\begin{aligned}
& +i\overline{u_{\alpha_1,i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_1}) \\
& +i\overline{\ell_{a_1,i_1}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_1}) \\
& +\frac{1}{2}iN_{A_1,r_1}{}^TC\gamma^{\mu_1}D_{\mu_1}(N_{A_1,r_1}) \\
& -\frac{1}{2}N_{A_1,r_1}{}^TCN_{A_1,r_1}M_F \\
& -\overline{q_{\alpha_1,a_1,i_1}}P_Rd_{\alpha_1,i_2}H_{a_1}Y_{d,i_1,i_2} \\
& -\overline{\ell_{a_1,i_1}}P_Re_{i_2}H_{a_1}Y_{e,i_1,i_2} \\
& +\overline{\ell_{a_1,i_1}}P_RN_{A_1,r_1}\text{NewScalar1}_{a_2}y_{1,i_1,r_1}\text{gen}(\text{SU2L(fund)})_{A_1,a_1,a_2} \\
& -\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_2}\bar{\epsilon}_{a_1,a_2} \\
& +\overline{y_{2,i_1,r_1}}N_{A_1,r_1}{}^TCPL\ell_{a_3,i_1}\text{NewScalar2}_{a_2}\text{gen}(\text{SU2L(fund)})_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_1}{}^\dagger \\
& -\overline{Y_{e,i_1,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}H_{a_1}{}^\dagger \\
& -\overline{q_{\alpha_1,a_2,i_1}}P_Ru_{\alpha_1,i_2}Y_{u,i_1,i_2}\epsilon_{a_2,a_1}H_{a_1}{}^\dagger \\
& +H_{a_1}\mu^2H_{a_1}{}^\dagger \\
& -\frac{1}{2}\lambda H_{a_1}H_{a_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger \\
& -M_{S_1}^2\text{NewScalar1}_{a_1}\text{NewScalar1}_{a_1}{}^\dagger \\
& +\overline{y_{1,i_1,r_1}}N_{A_1,r_1}{}^TCPL\ell_{a_2,i_1}\text{gen}(\text{SU2L(fund)})_{A_1,a_1,a_2}\text{NewScalar1}_{a_1}{}^\dagger \\
& +\text{lambdaH1Inv1}H_{a_3}\text{NewScalar1}_{a_4}\text{T3HiggsPortal1CGInv1}_{a_1,a_3,a_2,a_4}H_{a_1}{}^\dagger\text{NewScalar1}_{a_2}{}^\dagger \\
& +\text{lambdaH1Inv2}H_{a_3}\text{NewScalar1}_{a_4}\text{T3HiggsPortal1CGInv2}_{a_1,a_3,a_2,a_4}H_{a_1}{}^\dagger\text{NewScalar1}_{a_2}{}^\dagger \\
& +\frac{1}{2}\text{lambdaS1Inv1NewScalar1}_{a_3}\text{NewScalar1}_{a_4}\text{T3Scalar1SelfCGInv1}_{a_1,a_3,a_2,a_4}\text{NewScalar1}_{a_1}{}^\dagger\text{NewScalar1}_{a_2}{}^\dagger \\
& +\overline{\lambda_{T_3}}\text{NewScalar2}_{a_1}\epsilon_{a_2,a_3}H_{a_1}{}^\dagger H_{a_2}{}^\dagger\text{NewScalar1}_{a_3}{}^\dagger \\
& -M_{S_2}^2\text{NewScalar2}_{a_1}\text{NewScalar2}_{a_1}{}^\dagger \\
& +\overline{\ell_{a_3,i_1}}P_RN_{A_1,r_1}y_{2,i_1,r_1}\epsilon_{a_2,a_1}\text{gen}(\text{SU2L(fund)})_{A_1,a_3,a_2}\text{NewScalar2}_{a_1}{}^\dagger \\
& +H_{a_1}H_{a_2}\text{NewScalar1}_{a_3}\lambda_{T_3}\bar{\epsilon}_{a_2,a_3}\text{NewScalar2}_{a_1}{}^\dagger \\
& +\text{lambdaH2Inv1}H_{a_3}\text{NewScalar2}_{a_4}\text{T3HiggsPortal2CGInv1}_{a_1,a_3,a_2,a_4}H_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +\text{lambdaH2Inv2}H_{a_3}\text{NewScalar2}_{a_4}\text{T3HiggsPortal2CGInv2}_{a_1,a_3,a_2,a_4}H_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +\text{lambda12Inv1NewScalar1}_{a_3}\text{NewScalar2}_{a_4}\text{T3CrossScalarCGInv1}_{a_1,a_3,a_2,a_4}\text{NewScalar1}_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +\text{lambda12Inv2NewScalar1}_{a_3}\text{NewScalar2}_{a_4}\text{T3CrossScalarCGInv2}_{a_1,a_3,a_2,a_4}\text{NewScalar1}_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +\frac{1}{2}\text{lambdaS2Inv1NewScalar2}_{a_3}\text{NewScalar2}_{a_4}\text{T3Scalar2SelfCGInv1}_{a_1,a_3,a_2,a_4}\text{NewScalar2}_{a_1}{}^\dagger\text{NewScalar2}_{a_2}{}^\dagger \\
& +D_{\mu_1}(H_{a_1}){}^\dagger D_{\mu_1}(H_{a_1}) \\
& +D_{\mu_1}(\text{NewScalar1}_{a_1}){}^\dagger D_{\mu_1}(\text{NewScalar1}_{a_1}) \\
& +D_{\mu_1}(\text{NewScalar2}_{a_1}){}^\dagger D_{\mu_1}(\text{NewScalar2}_{a_1})
\end{aligned}$$

$$\text{Matched EFT Lagrangian} - \frac{3\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^4}{16\left(M_F^2-M_{S_1}^2\right)^2}$$

$$- \frac{3\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}M_F^4}{16\left(M_F^2-M_{S_2}^2\right)^2}$$

$$\begin{aligned}
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} \ell_{a_4,i_1}^T C P_L \ell_{a_1,i_2} H_{a_2} H_{a_3} M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{4 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} \ell_{a_4,i_1}^T C P_L \ell_{a_1,i_2} H_{a_2} H_{a_3} M_{S1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{4 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_2,a_4} \epsilon_{a_3,a_1} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{4 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T M_{S1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_2,a_4} \epsilon_{a_3,a_1} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{4 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{3h \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)} \\
& + \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)} \\
& + \frac{3h \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{3h \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S2}^2)} \\
& + \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S2}^2)} \\
& + \frac{3h \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{3h \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 (M_F^2 - M_{S1}^2)} \\
& + \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 (M_F^2 - M_{S1}^2)} \\
& + \frac{3h \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{16 (M_F^2 - M_{S1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 \left(M_F^2 - M_{S1}^2 \right)^2} \\
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 \left(M_F^2 - M_{S2}^2 \right)} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 \left(M_F^2 - M_{S2}^2 \right)} \\
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{16 \left(M_F^2 - M_{S2}^2 \right)^2} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 \left(M_F^2 - M_{S2}^2 \right)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_1}}^T C P_L \ell_{a_1,i_2} H_{a_2} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{4 \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_1}}^T C P_L \ell_{a_1,i_2} H_{a_2} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{4 \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_2,a_4} \epsilon_{a_3,a_1} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{4 \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_2,a_4} \epsilon_{a_3,a_1} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{4 \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{12\epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{\left(G_{\mu_1 \mu_2}^{X_1} \right)^2}{4g_s^2} \\
& - \frac{13\hbar \left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{12\epsilon} \\
& - \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2
\end{aligned}$$

$$\begin{aligned}
& -\frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2} \\
& +i\overline{d_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(d_{\alpha_1,i_2}) \\
& +i\overline{e_{i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(e_{i_2}) \\
& +i\overline{q_{\alpha_1,a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(q_{\alpha_1,a_1,i_2}) \\
& +i\overline{u_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_2}) \\
& +i\overline{\ell_{a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_2}) \\
& -\overline{q_{\alpha_1,a_1,i_2}}P_Rd_{\alpha_1,i_1}H_{a_1}Y_{d,i_2,i_1} \\
& -\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}Y_{e,i_2,i_1} \\
& +\frac{3h\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon} \\
& -\frac{3}{32}h\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{3h\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8(M_F^2-M_{S1}^2)} \\
& -\frac{3h\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8(M_F^2-M_{S1}^2)} \\
& -\frac{3h\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16(M_F^2-M_{S1}^2)^2} \\
& -\frac{3h\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16(M_F^2-M_{S1}^2)^2} \\
& +\frac{3h\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon} \\
& -\frac{3}{32}h\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{3h\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{8(M_F^2-M_{S2}^2)} \\
& -\frac{3h\log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right)\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{8(M_F^2-M_{S2}^2)} \\
& -\frac{3h\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{16(M_F^2-M_{S2}^2)^2} \\
& -\frac{3h\log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right)\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{16(M_F^2-M_{S2}^2)^2} \\
& -\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_2}\bar{e}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_1}^\dagger \\
& -\overline{Y_{e,i_1,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}H_{a_1}^\dagger \\
& +C_{H2}H_{a_1}H_{a_1}^\dagger \\
& +\frac{3h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{16\epsilon}
\end{aligned}$$

$$\begin{aligned}
& -\frac{3}{32}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger \\
& -\frac{3h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{8\left(M_F^2-M_{S1}^2\right)} \\
& -\frac{3h\log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{8\left(M_F^2-M_{S1}^2\right)} \\
& -\frac{3h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{16\left(M_F^2-M_{S1}^2\right)^2} \\
& -\frac{3h\log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{16\left(M_F^2-M_{S1}^2\right)^2} \\
& +\frac{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{16\epsilon} \\
& -\frac{3}{32}\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger \\
& -\frac{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S2}^2y_{2,i_3,r_1}H_{a_1}^\dagger}{8\left(M_F^2-M_{S2}^2\right)} \\
& -\frac{3h\log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S2}^2y_{2,i_3,r_1}H_{a_1}^\dagger}{8\left(M_F^2-M_{S2}^2\right)} \\
& -\frac{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S2}^4y_{2,i_3,r_1}H_{a_1}^\dagger}{16\left(M_F^2-M_{S2}^2\right)^2} \\
& -\frac{3h\log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S2}^4y_{2,i_3,r_1}H_{a_1}^\dagger}{16\left(M_F^2-M_{S2}^2\right)^2} \\
& -\overline{q_{\alpha_1,a_2,i_2}}P_Ru_{\alpha_1,i_1}Y_{u,i_2,i_1}\epsilon_{a_2,a_1}H_{a_1}^\dagger \\
& -\frac{1}{2}\lambda H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{\text{lambdaH1Inv1}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{5\text{lambdaH1Inv2}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& +\frac{\text{lambdaH2Inv1}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{5\text{lambdaH2Inv2}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& +\frac{\text{lambdaH1Inv1lambdaH1Inv2}hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& +\frac{\text{lambdaH2Inv1lambdaH2Inv2}hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& +\frac{1}{2}\text{lambdaH1Inv1}^2h\log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{5}{6}\text{lambdaH1Inv2}^2h\log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \frac{\text{lambdaH1Inv1}\text{lambdaH1Inv2}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{1}{2} \text{lambdaH2Inv1}^2 \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6} \text{lambdaH2Inv2}^2 \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{\hbar \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{\hbar \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}1}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S}1}^2 - M_{\text{S}2}^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2}\right) \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}1}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S}1}^2 - M_{\text{S}2}^2} \\
& - \frac{\hbar \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}2}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S}1}^2 - M_{\text{S}2}^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2}\right) \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}2}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S}1}^2 - M_{\text{S}2}^2} \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2}
\end{aligned}$$

$$T3 - A, \quad \alpha = 0 : \quad S_1 = (1, 0), \quad S_2 = (3, 1), \quad F = (2, \tfrac{1}{2})$$

$$\begin{aligned}
& \text{Full UV Lagrangian} - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2} - \frac{\left(G_{\mu_1 \mu_2}^{X_1}\right)^2}{4g_s^2} \\
& - \frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4g_L^2} \\
& + \frac{1}{2} \text{lambdaS1Inv1NewScalar1}^2 \left(\text{NewScalar1}^\dagger\right)^2 \\
& + i \overline{d_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1, i_1}) \\
& + i \overline{e_{i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_1}) \\
& + i \overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1}) \\
& + i \overline{q_{\alpha_1, a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1, a_1, i_1}) \\
& + i \overline{u_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1, i_1}) \\
& + i \overline{\ell_{a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1, i_1}) \\
& - \overline{N_{a_1, r_1}} N_{a_1, r_1} M_F \\
& - \overline{q_{\alpha_1, a_1, i_1}} P_R d_{\alpha_1, i_2} H_{a_1} Y_{d, i_1, i_2}
\end{aligned}$$

$$\begin{aligned}
& -\overline{\ell_{a_1, i_1}} P_R e_{i_2} H_{a_1} Y_{e, i_1, i_2} \\
& + \text{NewScalar1} \overline{\ell_{a_1, i_1}} C P_R \overline{N_{a_2, r_1}}^T y_{1, i_1, r_1} \epsilon_{a_1, a_2} \\
& + \overline{y_{2, i_1, r_1}} \overline{N_{a_1, r_1}} P_L \ell_{a_2, i_1} \text{NewScalar2}_{A_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \\
& - \overline{Y_{u, i_1, i_2}} \overline{u_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_2} \bar{e}_{a_1, a_2} \\
& - \text{NewScalar1} M_{S_1}^2 \text{NewScalar1}^\dagger \\
& + \overline{y_{1, i_1, r_1}} \ell_{a_1, i_1}^T C P_L N_{a_2, r_1} \bar{e}_{a_1, a_2} \text{NewScalar1}^\dagger \\
& - \overline{Y_{d, i_1, i_2}} \overline{d_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_1}^\dagger \\
& - \overline{Y_{e, i_1, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} H_{a_1}^\dagger \\
& - \overline{q_{\alpha_1, a_2, i_1}} P_R u_{\alpha_1, i_2} Y_{u, i_1, i_2} \epsilon_{a_2, a_1} H_{a_1}^\dagger \\
& + \text{lambdaH1Inv1NewScalar1} H_{a_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2} \text{NewScalar1}^\dagger H_{a_1}^\dagger \\
& + H_{a_1} \mu^2 H_{a_1}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \overline{\lambda_{\text{T3}}} \text{NewScalar2}_{A_1} \epsilon_{a_3, a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3} \text{NewScalar1}^\dagger H_{a_1}^\dagger H_{a_2}^\dagger \\
& - M_{S_2}^2 \text{NewScalar2}_{A_1} \text{NewScalar2}_{A_1}^\dagger \\
& + \overline{\ell_{a_1, i_1}} P_R N_{a_2, r_1} y_{2, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \text{NewScalar2}_{A_1}^\dagger \\
& + \text{NewScalar1} H_{a_2} H_{a_3} \lambda_{\text{T3}} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3} \bar{e}_{a_1, a_2} \text{NewScalar2}_{A_1}^\dagger \\
& + \text{lambda12Inv1NewScalar1NewScalar2}_{A_2} \text{T3CrossScalarCGInv1}_{A_1, A_2} \text{NewScalar1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv1NewScalar2}_{A_3} \text{NewScalar2}_{A_4} \text{T3Scalar2SelfCGInv1}_{A_1, A_3, A_2, A_4} \text{NewScalar2}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv2NewScalar2}_{A_3} \text{NewScalar2}_{A_4} \text{T3Scalar2SelfCGInv2}_{A_1, A_3, A_2, A_4} \text{NewScalar2}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + D_{\mu_1} (\text{NewScalar1})^\dagger D_{\mu_1} (\text{NewScalar1}) \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& + D_{\mu_1} (\text{NewScalar2}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{A_1}) \\
& \text{Matched EFT Lagrangian} - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 \left(M_F^2 - M_{S_1}^2 \right)^2} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{16 \left(M_F^2 - M_{S_2}^2 \right)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^4}{16 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)} \\
& + \frac{3\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)} \\
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{16(M_F^2 - M_{S2}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{1}{4} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{\left(G_{\mu_1\mu_2}^{X_1}\right)^2}{4g_s^2} \\
& - \frac{2\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& - \frac{1}{6} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) \left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& - \frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2} \\
& + \overline{id_{\alpha_1,i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1,i_2}) \\
& + i \overline{e_{i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_2}) \\
& + i \overline{q_{\alpha_1,a_1,i_2}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1,a_1,i_2}) \\
& + i \overline{u_{\alpha_1,i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1,i_2})
\end{aligned}$$

$$\begin{aligned}
& + \overline{i\ell_{a_1,i_2}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1,i_2}) \\
& - \overline{q_{\alpha_1,a_1,i_2}} P_R d_{\alpha_1,i_1} H_{a_1} Y_{d,i_2,i_1} \\
& - \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} Y_{e,i_2,i_1} \\
& + \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{3\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16\epsilon} \\
& - \frac{3}{32} \hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{8(M_F^2 - M_{S2}^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{8(M_F^2 - M_{S2}^2)} \\
& - \frac{3\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16(M_F^2 - M_{S2}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16(M_F^2 - M_{S2}^2)^2} \\
& - \overline{Y_{u,i_1,i_2}} \overline{u_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_2} \bar{e}_{a_1,a_2} \\
& - \overline{Y_{d,i_1,i_2}} \overline{d_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_1}^\dagger \\
& - \overline{Y_{e,i_1,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} H_{a_1}^\dagger \\
& + \text{CH21} H_{a_1} H_{a_1}^\dagger \\
& + \hbar C_{H2} H_{a_2} \text{T3HiggsPortal1CGInv1}_{a_1,a_2} H_{a_1}^\dagger \\
& + \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S1}^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}1}^2 y_{1,i_3,r_1} H_{a_1}{}^\dagger}{2 (M_F^2 - M_{\text{S}1}^2)} \\
& - \frac{\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}1}^4 y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4 (M_F^2 - M_{\text{S}1}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}1}^4 y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4 (M_F^2 - M_{\text{S}1}^2)^2} \\
& + \frac{3 \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{16 \epsilon} \\
& - \frac{\frac{3}{32} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{3 \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^2 y_{2,i_3,r_1} H_{a_1}{}^\dagger} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^2 y_{2,i_3,r_1} H_{a_1}{}^\dagger}{8 (M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^2 y_{2,i_3,r_1} H_{a_1}{}^\dagger}{8 (M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{3 \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^4 y_{2,i_3,r_1} H_{a_1}{}^\dagger}{16 (M_F^2 - M_{\text{S}2}^2)^2} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^4 y_{2,i_3,r_1} H_{a_1}{}^\dagger}{16 (M_F^2 - M_{\text{S}2}^2)^2} \\
& - \overline{q_{\alpha_1,a_2,i_2}} P_R u_{\alpha_1,i_1} Y_{u,i_2,i_1} \epsilon_{a_2,a_1} H_{a_1}{}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger \\
& + \frac{3 \text{lambdaH2Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2 \epsilon} \\
& + \frac{3}{2} \text{lambdaH2Inv}2^2 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger \\
& + \frac{\text{lambdaH1Inv}1^2 \hbar H_{a_3} H_{a_4} \text{T3HiggsPortal1CGInv}1_{a_1,a_3} \text{T3HiggsPortal1CGInv}1_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2 \epsilon} \\
& + \frac{1}{2} \text{lambdaH1Inv}1^2 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2} \right) H_{a_3} H_{a_4} \text{T3HiggsPortal1CGInv}1_{a_1,a_3} \text{T3HiggsPortal1CGInv}1_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger \\
& + \frac{\hbar \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} \lambda_{\text{T}3} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2 \epsilon} \\
& + \frac{\hbar \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}1}^2 \lambda_{\text{T}3} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2 (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2} \right) \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}1}^2 \lambda_{\text{T}3} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2 (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& - \frac{\hbar \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}2}^2 \lambda_{\text{T}3} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2 (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}2}^2 \lambda_{\text{T}3} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2 (M_{\text{S}1}^2 - M_{\text{S}2}^2)}
\end{aligned}$$

$$+D_{\mu_1}(H_{a_1})^\dagger D_{\mu_1}(H_{a_1})$$

$$-\frac{B_{\mu_1\mu_2}^2}{4g_Y^2}$$

$$T3-D, \alpha=-2: \quad S_1=(3,-1), \quad S_2=(1,0), \quad F=(2,-\frac{1}{2})$$

$$\text{Full UV Lagrangian} -\frac{B_{\mu_1\mu_2}^2}{4g_Y^2} -\frac{\left(G_{\mu_1\mu_2}^{X_1}\right)^2}{4g_s^2}$$

$$-\frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2}$$

$$+\frac{1}{2}\text{lambdaS2Inv1NewScalar2}^2\left(\text{NewScalar2}^\dagger\right)^2$$

$$+i\overline{d_{\alpha_1,i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(d_{\alpha_1,i_1})$$

$$+i\overline{e_{i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(e_{i_1})$$

$$+\text{NewScalar2}\overline{y_{2,i_1,r_1}}\overline{N_{a_1,r_1}}P_L\ell_{a_1,i_1}$$

$$+i\overline{N_{a_1,r_1}}\gamma^{\mu_1}D_{\mu_1}(N_{a_1,r_1})$$

$$+i\overline{q_{\alpha_1,a_1,i_1}}\gamma^{\mu_1}P_LD_{\mu_1}(q_{\alpha_1,a_1,i_1})$$

$$+i\overline{u_{\alpha_1,i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_1})$$

$$+i\overline{\ell_{a_1,i_1}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_1})$$

$$-\overline{N_{a_1,r_1}}N_{a_1,r_1}M_F$$

$$-\overline{q_{\alpha_1,a_1,i_1}}P_Rd_{\alpha_1,i_2}H_{a_1}Y_{d,i_1,i_2}$$

$$-\overline{\ell_{a_1,i_1}}P_Re_{i_2}H_{a_1}Y_{e,i_1,i_2}$$

$$+\overline{\ell_{a_3,i_1}}CP_R\overline{N_{a_1,r_1}}^T\text{NewScalar1}_{A_1}y_{1,i_1,r_1}\epsilon_{a_1,a_2}\text{gen}(\text{SU2L(fund)})_{A_1,a_3,a_2}$$

$$-\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_2}\bar{\epsilon}_{a_1,a_2}$$

$$-\text{NewScalar2}M_{S_2}^2\text{NewScalar2}^\dagger$$

$$+\overline{\ell_{a_1,i_1}}P_RN_{a_1,r_1}y_{2,i_1,r_1}\text{NewScalar2}^\dagger$$

$$+H_{a_2}H_{a_3}\text{NewScalar1}_{A_1}\lambda_{\text{T3}}\text{gen}(\text{SU2L(fund)})_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar2}^\dagger$$

$$-\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_1}^\dagger$$

$$-\overline{Y_{e,i_1,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}H_{a_1}^\dagger$$

$$-\overline{q_{\alpha_1,a_2,i_1}}P_Ru_{\alpha_1,i_2}Y_{u,i_1,i_2}\epsilon_{a_2,a_1}H_{a_1}^\dagger$$

$$+\text{lambdaH2Inv1NewScalar2}H_{a_2}\text{T3HiggsPortal2CGInv1}_{a_1,a_2}\text{NewScalar2}^\dagger H_{a_1}^\dagger$$

$$+H_{a_1}\mu^2H_{a_1}^\dagger$$

$$-\frac{1}{2}\lambda H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$$

$$-M_{S_1}^2\text{NewScalar1}_{A_1}\text{NewScalar1}_{A_1}^\dagger$$

$$+\overline{y_{1,i_1,r_1}}\overline{\ell_{a_3,i_1}}^TP_LN_{a_1,r_1}\text{gen}(\text{SU2L(fund)})_{A_1,a_2,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar1}_{A_1}^\dagger$$

$$+\text{lambda12Inv1NewScalar2NewScalar1}_{A_2}\text{T3CrossScalarCGInv1}_{A_1,A_2}\text{NewScalar2}^\dagger\text{NewScalar1}_{A_1}^\dagger$$

$$+\frac{1}{2}\text{lambdaH1Inv1}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$$

$$+\frac{1}{2}\text{lambdaH1Inv1}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,A_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$$

$$\begin{aligned}
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \text{NewScalar2} \overline{\lambda_{\text{T3}} \epsilon_{a_3, a_2} \text{gen}(\text{SU2L}(\text{fund}))}_{A_1, a_1, a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{A_3} \text{NewScalar1}_{A_4} \text{T3Scalar1SelfCGInv1}_{A_1, A_3, A_2, A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar1}_{A_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv2} \text{NewScalar1}_{A_3} \text{NewScalar1}_{A_4} \text{T3Scalar1SelfCGInv2}_{A_1, A_3, A_2, A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar1}_{A_2}^\dagger \\
& + D_{\mu_1} (\text{NewScalar2})^\dagger D_{\mu_1} (\text{NewScalar2}) \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& + D_{\mu_1} (\text{NewScalar1}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{A_1}) \\
& \text{Matched EFT Lagrangian} - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{16 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} \overline{\ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S2}^2 \lambda_{\text{T3}} \bar{\epsilon}_{a_1, a_2} \bar{\epsilon}_{a_3, a_4} M_F^3}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} \overline{\ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 \lambda_{\text{T3}} \bar{\epsilon}_{a_1, a_2} \bar{\epsilon}_{a_3, a_4} M_F^3}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{\lambda_{\text{T3}} \ell_{a_3, i_1}} C P_R \overline{\ell_{a_4, i_2}}^T M_{S2}^2 y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{\lambda_{\text{T3}} \ell_{a_3, i_1}} C P_R \overline{\ell_{a_4, i_2}}^T M_{S1}^2 y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{3\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S_1}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S_1}^2)} \\
& + \frac{3\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{16(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}^T} C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}^T} C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}^T} M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}^T} M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{3\hbar B_{\mu_1 \mu_2}^2}{4\epsilon}
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right)B_{\mu_1\mu_2}^2 \\
& -\frac{1}{4}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)B_{\mu_1\mu_2}^2 \\
& -\frac{\left(G_{\mu_1\mu_2}^{X_1}\right)^2}{4g_s^2} \\
& -\frac{2\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon} \\
& -\frac{1}{2}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{1}{6}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2} \\
& +i\overline{d_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(d_{\alpha_1,i_2}) \\
& +i\overline{e_{i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(e_{i_2}) \\
& +i\overline{q_{\alpha_1,a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(q_{\alpha_1,a_1,i_2}) \\
& +i\overline{u_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_2}) \\
& +i\overline{\ell_{a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_2}) \\
& -\overline{q_{\alpha_1,a_1,i_2}}P_Rd_{\alpha_1,i_1}H_{a_1}Y_{d,i_2,i_1} \\
& -\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}Y_{e,i_2,i_1} \\
& +\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8(M_F^2-M_{S1}^2)} \\
& -\frac{3\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8(M_F^2-M_{S1}^2)} \\
& -\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16(M_F^2-M_{S1}^2)^2} \\
& -\frac{3\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16(M_F^2-M_{S1}^2)^2} \\
& +\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\
& -\frac{1}{8}\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{2(M_F^2-M_{S2}^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 (M_F^2 - M_{S_2}^2)} \\
& - \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 (M_F^2 - M_{S_2}^2)^2} \\
& - \overline{Y_{u,i_1,i_2} u_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_2} \bar{e}_{a_1,a_2} \\
& - \overline{Y_{d,i_1,i_2} d_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_1}^\dagger \\
& - \overline{Y_{e,i_1,i_2} e_{i_2}} P_L \ell_{a_1,i_1} H_{a_1}^\dagger \\
& + C_{H2} H_{a_1} H_{a_1}^\dagger \\
& + \text{CH21} h H_{a_2} \text{T3HiggsPortal2CGInv1}_{a_1,a_2} H_{a_1}^\dagger \\
& + \frac{3 \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{16 \epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3 \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{8 (M_F^2 - M_{S_1}^2)} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{8 (M_F^2 - M_{S_1}^2)} \\
& - \frac{3 \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{S_1}^2)^2} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4 \epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 (M_F^2 - M_{S_2}^2)} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 (M_F^2 - M_{S_2}^2)} \\
& - \frac{\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 (M_F^2 - M_{S_2}^2)^2} \\
& - \overline{q_{\alpha_1,a_2,i_2}} P_R u_{\alpha_1,i_1} Y_{u,i_2,i_1} \epsilon_{a_2,a_1} H_{a_1}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \frac{3\text{lambdaH1Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv}2^2 \hbar \log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH2Inv}1^2 \hbar H_{a_3} H_{a_4} \text{T3HiggsPortal2CGInv}1_{a_1, a_3} \text{T3HiggsPortal2CGInv}1_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \text{lambdaH2Inv}1^2 \hbar \log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right) H_{a_3} H_{a_4} \text{T3HiggsPortal2CGInv}1_{a_1, a_3} \text{T3HiggsPortal2CGInv}1_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S_1}^2 - M_{S_2}^2)} \\
& + D_{\mu_1}(H_{a_1})^\dagger D_{\mu_1}(H_{a_1}) \\
& - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2}
\end{aligned}$$

$$T3 - E, \quad \alpha = 0 : \quad S_1 = (3, 0), \quad S_2 = (3, 1), \quad F = (2, \tfrac{1}{2})$$

$$\text{Full UV Lagrangian} - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2} - \frac{(G_{\mu_1 \mu_2}^{X_1})^2}{4g_s^2}$$

$$\begin{aligned}
& - \frac{(W_{\mu_1 \mu_2}^{A_1})^2}{4g_L^2} \\
& + i \overline{d_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1, i_1}) \\
& + i \overline{e_{i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_1}) \\
& + i \overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1}) \\
& + i \overline{q_{\alpha_1, a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1, a_1, i_1}) \\
& + i \overline{u_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1, i_1}) \\
& + i \overline{\ell_{a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1, i_1}) \\
& - \overline{N_{a_1, r_1}} N_{a_1, r_1} M_F \\
& - \overline{q_{\alpha_1, a_1, i_1}} P_R d_{\alpha_1, i_2} H_{a_1} Y_{d, i_1, i_2} \\
& - \overline{\ell_{a_1, i_1}} P_R e_{i_2} H_{a_1} Y_{e, i_1, i_2} \\
& + \overline{y_{2, i_1, r_1}} \overline{N_{a_1, r_1}} P_L \ell_{a_2, i_1} \text{NewScalar}_{2A_1 \text{gen}(\text{SU2L(fund)})}_{A_1, a_1, a_2} \\
& + \overline{\ell_{a_3, i_1}} C P_R \overline{N_{a_1, r_1}}^T \text{NewScalar}_{1A_1 y_{1, i_1, r_1} \epsilon_{a_1, a_2} \text{gen}(\text{SU2L(fund)})}_{A_1, a_3, a_2}
\end{aligned}$$

$$\begin{aligned}
& -\overline{Y_{u,i_1,i_2}} \overline{u_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_2} \bar{e}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}} \overline{d_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_1}^\dagger \\
& -\overline{Y_{e,i_1,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} H_{a_1}^\dagger \\
& -\overline{q_{\alpha_1,a_2,i_1}} P_R u_{\alpha_1,i_2} Y_{u,i_1,i_2} e_{a_2,a_1} H_{a_1}^\dagger \\
& +H_{a_1} \mu^2 H_{a_1}^\dagger \\
& -\frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& -M_{S1}^2 \text{NewScalar1}_{A_1} \text{NewScalar1}_{A_1}^\dagger \\
& +\overline{y_{1,i_1,r_1}} \ell_{a_3,i_1}^T C P_L N_{a_1,r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_2,a_3} \bar{e}_{a_1,a_2} \text{NewScalar1}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv1}_{a_1,a_2,A_1,A_2} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv1}_{a_1,a_2,A_2,A_1} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv2}_{a_1,a_2,A_1,A_2} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv2}_{a_1,a_2,A_2,A_1} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& +\overline{\lambda_{T3}} \text{NewScalar2}_{A_2} e_{a_3,a_2} \text{fStruct}(\text{SU2L})_{A_1,A_2,A_3} \text{gen}(\text{SU2L}(\text{fund}))_{A_3,a_1,a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{A_3} \text{NewScalar1}_{A_4} \text{T3Scalar1SelfCGInv1}_{A_1,A_3,A_2,A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar1}_{A_2}^\dagger \\
& +\frac{1}{2} \text{lambdaS1Inv2} \text{NewScalar1}_{A_3} \text{NewScalar1}_{A_4} \text{T3Scalar1SelfCGInv2}_{A_1,A_3,A_2,A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar1}_{A_2}^\dagger \\
& -M_{S2}^2 \text{NewScalar2}_{A_1} \text{NewScalar2}_{A_1}^\dagger \\
& +\overline{\ell_{a_1,i_1}} P_R N_{a_2,r_1} y_{2,i_1,r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_2} \text{NewScalar2}_{A_1}^\dagger \\
& -H_{a_2} H_{a_3} \text{NewScalar1}_{A_2} \lambda_{T3} \text{fStruct}(\text{SU2L})_{A_1,A_2,A_3} \text{gen}(\text{SU2L}(\text{fund}))_{A_3,a_1,a_3} \bar{e}_{a_1,a_2} \text{NewScalar2}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv1}_{a_1,a_2,A_1,A_2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv1}_{a_1,a_2,A_2,A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv2}_{a_1,a_2,A_1,A_2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& +\frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv2}_{a_1,a_2,A_2,A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& +\text{lambda12Inv1} \text{NewScalar1}_{A_3} \text{NewScalar2}_{A_4} \text{T3CrossScalarCGInv1}_{A_1,A_3,A_2,A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& +\text{lambda12Inv2} \text{NewScalar1}_{A_3} \text{NewScalar2}_{A_4} \text{T3CrossScalarCGInv2}_{A_1,A_3,A_2,A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& +\text{lambda12Inv3} \text{NewScalar1}_{A_3} \text{NewScalar2}_{A_4} \text{T3CrossScalarCGInv3}_{A_1,A_3,A_2,A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& +\frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{A_3} \text{NewScalar2}_{A_4} \text{T3Scalar2SelfCGInv1}_{A_1,A_3,A_2,A_4} \text{NewScalar2}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& +\frac{1}{2} \text{lambdaS2Inv2} \text{NewScalar2}_{A_3} \text{NewScalar2}_{A_4} \text{T3Scalar2SelfCGInv2}_{A_1,A_3,A_2,A_4} \text{NewScalar2}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& +D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& +D_{\mu_1} (\text{NewScalar1}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{A_1}) \\
& +D_{\mu_1} (\text{NewScalar2}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{A_1})
\end{aligned}$$

$$\text{Matched EFT Lagrangian} - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{16 (M_F^2 - M_{S1}^2)^2}$$

$$\begin{aligned}
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{16 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{S_1}^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{i\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{i\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{i\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{i\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)} \\
& + \frac{3\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 (M_F^2 - M_{S_1}^2)}
\end{aligned}$$

$$\begin{aligned}
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8(M_F^2 - M_{S1}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{16(M_F^2 - M_{S1}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8(M_F^2 - M_{S1}^2)^2} \\
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)} \\
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{16(M_F^2 - M_{S2}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)^2} \\
& - \frac{i\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} \ell_{a_4,i_2}{}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{i\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} \ell_{a_4,i_2}{}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{i\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \bar{\ell}_{a_4,i_2}{}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{i\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \bar{\ell}_{a_4,i_2}{}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{1}{4} \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{(G_{\mu_1\mu_2}^{X_1})^2}{4g_s^2} \\
& - \frac{5\hbar (W_{\mu_1\mu_2}^{A_1})^2}{6\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) (W_{\mu_1\mu_2}^{A_1})^2
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{\text{S}1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{\text{S}2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2} \\
& +i\overline{d_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(d_{\alpha_1,i_2}) \\
& +i\overline{e_{i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(e_{i_2}) \\
& +i\overline{q_{\alpha_1,a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(q_{\alpha_1,a_1,i_2}) \\
& +i\overline{u_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_2}) \\
& +i\overline{\ell_{a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_2}) \\
& -\overline{q_{\alpha_1,a_1,i_2}}P_Rd_{\alpha_1,i_1}H_{a_1}Y_{d,i_2,i_1} \\
& -\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}Y_{e,i_2,i_1} \\
& +\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{\text{S}1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8\left(M_F^2-M_{\text{S}1}^2\right)} \\
& -\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_{\text{S}1}^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{\text{S}1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8\left(M_F^2-M_{\text{S}1}^2\right)} \\
& -\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{\text{S}1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16\left(M_F^2-M_{\text{S}1}^2\right)^2} \\
& -\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_{\text{S}1}^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{\text{S}1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16\left(M_F^2-M_{\text{S}1}^2\right)^2} \\
& +\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{\text{S}2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{8\left(M_F^2-M_{\text{S}2}^2\right)} \\
& -\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_{\text{S}2}^2}\right)\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{\text{S}2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{8\left(M_F^2-M_{\text{S}2}^2\right)} \\
& -\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{\text{S}2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{16\left(M_F^2-M_{\text{S}2}^2\right)^2} \\
& -\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_{\text{S}2}^2}\right)\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{\text{S}2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{16\left(M_F^2-M_{\text{S}2}^2\right)^2} \\
& -\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_2}\bar{e}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_1}^\dagger
\end{aligned}$$

$$\begin{aligned}
& -\overline{Y_{e,i_1,i_2} e_{i_2}} P_L \ell_{a_1,i_1} H_{a_1}^\dagger \\
& + C_{H2} H_{a_1} H_{a_1}^\dagger \\
& + \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{8(M_F^2 - M_{S1}^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{8(M_F^2 - M_{S1}^2)} \\
& - \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16(M_F^2 - M_{S1}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16(M_F^2 - M_{S1}^2)^2} \\
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{8(M_F^2 - M_{S2}^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{8(M_F^2 - M_{S2}^2)} \\
& - \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{16(M_F^2 - M_{S2}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{16(M_F^2 - M_{S2}^2)^2} \\
& - \overline{q_{\alpha_1,a_2,i_2}} P_R u_{\alpha_1,i_1} Y_{u,i_2,i_1} \epsilon_{a_2,a_1} H_{a_1}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{3\text{lambdaH1Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3\text{lambdaH2Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv}2^2 \hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{3}{2} \text{lambdaH2Inv}2^2 \hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}1}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S}1}^2 - M_{\text{S}2}^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2}\right) \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}1}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S}1}^2 - M_{\text{S}2}^2} \\
& - \frac{\hbar \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}2}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S}1}^2 - M_{\text{S}2}^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2}\right) \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}2}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S}1}^2 - M_{\text{S}2}^2} \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2}
\end{aligned}$$
